

STATEMENT OF PURPOSE

Krishna Priya, Rakhi, MS in data science, summer 2025

University of Mannheim

“Those who rule data will rule the entire world.” - Masayoshi Son.

I intend to develop a focus on acquiring a postgraduate degree in data science since I have recognized the promise therein almost naturally, as a scholar who carried out substantial qualitative research in related fields. Data science is the study of the extraction of knowledge from massive amounts of data, thereof analyzed by various scientific techniques, methods, and algorithms. Its i.e., data science importance in contemporaneous societies is found to rise faster than of late. In short, this calls for hasty growth of this industry, and what is influencing it is data transformation. It made me such a studious researcher about the domain as I observed all along and concluded that data science could be used in almost every field, a brief listing of which might start with agriculture, human resources, information technology, and every single industry involving commercial activities. Data science is a rapidly evolving field with several applications in various industries. I will be able to explore a variety of career alternatives that capitalize on my key abilities by combining my education in computer science engineering with my potential data science competence.

Obtaining a solid education has been essential for success in the job market since I was in school. To follow my passions, I have always placed a great value on education and skill development. My academic journey reflects a consistent growth in computer science. I achieved notable academic success in my early education, attaining a **10 CGPA** in 10th grade and **82.8%** in 12th grade from IES Public School, Thrissur. I earned my **Bachelor's in technology in Computer Science and Engineering** from APJ Abdul Kalam Technological University, graduating with a **CGPA of 9.02 and securing** the top position in my college. I was honored as the best outgoing student for the 2020-24 academic years. My impactful projects during my bachelor's studies greatly fueled my enthusiasm for technology. I received an S grade for my final year Research-oriented project on '**Indian Sign Language Recognition: Multi-dataset Approach**', showcasing deep learning systems and highlighting technology's potential for positive change. My project on 'Cardiovascular Disease Prediction using Machine Learning' further solidified my passion for AI and data science, earning commendations from my mentors during my **internship at the ICT Academy of Kerala**. Various internships have significantly contributed to my professional development. The 5-month 'Data Science with Python' training under Datapy Company enriched my skills in Data Manipulation and Analysis.

During my bachelor's program, I developed a strong interest in Deep Learning, earning an S grade in the subject, along with A+ grades in Artificial Intelligence and Machine Learning, solidifying my technical foundation. My passion for data structures, algorithms, and programming languages like C, Java, and Python has honed my problem-solving skills. I also earned top grades in practical labs, including Java, System Software, and Microprocessor Labs, and completed projects like a Mental Health Tracking System using Full-stack Development. These experiences have deepened my practical knowledge and prepared me to apply theoretical concepts in real-world scenarios.

Alongside my academics, I actively participated in hackathons, workshops, sports meets, and the college Arts Festival, contributing to my personal growth. As a member of the Innovation and Entrepreneurship Development Centre (IEDC) and IEEE Student Branch, I organized events like Python workshops and boot camps. Placement offers from Datapy and Qspiders broadened my perspective and boosted my confidence in my skills and professional competence.

Applying for a Master's degree abroad will enable me to continue my academic route; Germany was chosen because of its outstanding reputation in world education. Germany values education most particularly in science and technology, hence I would want to study there. The first will be the great teaching and always shifting courses fulfilling global standards. Surrounded in an environment that accepts cultural variety and provides international students with a safe place make me happy. Accepting the variety and rich German legacy will enable me to improve my learning process. Germany's educational system will support both personally and professionally development. Studying in a country known for its varied historical background, tolerance, and kind attitude can help overseas students shine here.

The University of Mannheim stands out for its global educational leadership, offering a highly ranked, internationally accredited Master's in Data Science. I am drawn to the program's focus on applying machine learning techniques to various data types and its practical projects in collaboration with industry partners, providing valuable exposure. I am eager to research under Prof. Dr.-Ing. Margret Keuper, focusing on the robustness and reliability of deep learning models and neural architecture search. The university's commitment to experiential learning aligns with my academic and professional goals, and I am confident I will be an asset to the program.

After completed my studies, I intend to enhance India's tech sector as a professional. With the IT industry set to grow significantly, there are numerous opportunities. My international education will provide me with a competitive advantage in roles such as **AI Engineer or Data Scientist at leading firms**. I am eager to support government projects like Digital India and Smart Cities Mission, utilizing my skills in IoT and AI. The knowledge acquired abroad will be crucial as India aims to become a global AI leader, with a significant proportion of AI professionals expected to be Indian. This global experience, coupled with my Indian heritage, will enable me to integrate global standards with local innovation.

As a dedicated scholar, my main goal is to join a prestigious organization and contribute to my personal development. I am committed to acquiring knowledge and skills to enhance my future career's effectiveness and reputation. My contributions can significantly boost productivity and success within any organization. This dedication to continuous learning and professional advancement also applies to my academic endeavors. I aim to bring the same enthusiasm and commitment to the university I will attend. My goal is to enrich the educational atmosphere and develop skills that will aid my future career, like my previous academic achievements through active engagement and collaboration. Once I achieve my professional milestones, I plan to start a socially responsible startup that creates opportunities and encourages innovation. Through this venture, I aim to contribute to charitable causes and support initiatives that address critical issues like poverty, education, and healthcare.

Thank you,

Krishna Priya, Rakhi